

AMATH 516: Homework 3
Sid Meka

1. **Exercise 3.2:** Let $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ be a convex function. Show that if there exists a point $x \in \text{ri}(\text{dom } f)$ with $f(x)$ finite, then f must be proper.

In order to show f is a proper convex function, we must have that $f(x) > -\infty$ for all x ; there exists an x , where $f(x) < \infty$; and $\text{dom } f \neq \emptyset$. By definition of the relative interior, for any $y \in \text{ri}(\text{dom } f)$, there exists an open ball, $B_r(x)$, around y that is completely contained within $\text{dom } f$. Furthermore, since $f(x) < \infty$ for some $x \in \text{ri}(\text{dom } f)$, and f is convex, it is not identically ∞ on $\text{dom } f$. Because $x \in \text{ri}(\text{dom } f)$, $x \in \text{dom } f$ by the definition of relative interior. Because $x \in \text{dom } f$, we see that $\text{dom } f$ is obviously not empty. Thus, $\text{dom } f \neq \emptyset$.

Now, we want to show that $f(x) > -\infty$ for all $x \in \mathbb{E}$. Suppose, for contradiction, that $f(y) = -\infty$ for some $y \in \mathbb{E}$. Then, for any $\lambda \in (0, 1)$, convexity gives:

$$\begin{aligned} f(\lambda x + (1 - \lambda)y) &\leq \lambda f(x) + (1 - \lambda)f(y) \\ &= \lambda f(x) + (1 - \lambda)(-\infty) \\ &= \lambda f(x) + \xi(-\infty) \text{ where } \xi \in (0, 1) \text{ from how we defined } \lambda \\ &= -\infty \end{aligned}$$

So $f(x) = -\infty$ for all points z on the line segment between x and y , which lie in $\text{dom } f$ since our open ball $B_r(x)$ is contained in the domain. This implies that $f \equiv -\infty$ on an open set. However, this contradicts the fact that $f(x) < \infty$ for some x , so such a y cannot exist. Therefore, $f(x) > -\infty$ for all $x \in \mathbb{E}$.

Conclusion: Because $f(x) > -\infty$ for all x ; there exists an x , where $f(x) < \infty$; and $\text{dom } f \neq \emptyset$, we have that f is a proper function.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

Let $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ be a convex function. Suppose $x \in \text{ri}(\text{dom } f)$ and $f(x) < \infty$. We will show that f is proper.

By definition of properness, we must verify that:

- $\text{dom}(f) \neq \emptyset$
- $f(x) > -\infty$ for all $x \in \text{dom}(f)$
- $f(x) < \infty$ for at least one $x \in \text{dom}(f)$

Since $x \in \text{ri}(\text{dom } f)$, and the relative interior is always nonempty for a convex domain, it follows that $\text{dom}(f) \neq \emptyset$.

Now suppose there exists $y \in \text{dom}(f)$ such that $f(y) = -\infty$. Then for any $\lambda \in (0, 1)$, convexity implies:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) = \lambda f(x) + (1 - \lambda)(-\infty) = -\infty.$$

Thus, $f \equiv -\infty$ on a segment inside the domain. Since convex functions that are finite at one point cannot equal $-\infty$ on any nontrivial interval, this contradicts convexity.

Therefore, $f(x) > -\infty$ for all $x \in \text{dom}(f)$, and since $f(x) < \infty$, the function is proper.

Here, the problem is that ChatGPT assumes that the relative interior is always nonempty for a convex domain when this is not the case. This is a mistake.

2. **Exercise 3.3:** Let $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ be a proper convex function. Show that the inequality

$$f\left(\sum_{i=1}^k \lambda_i x_i\right) \leq \sum_{i=1}^k \lambda_i f(x_i)$$

holds for any integer $k \in \mathbb{N}$, points $x_1, \dots, x_k \in \mathbb{E}$, and weights $\lambda \in \Delta_k$.

Case $k = 0$: There are no points x_i and no weights λ_i , so Δ_0 is empty. Hence, when $k = 0$, the hypothesis is never satisfied and the inequality is vacuously true. We proceed by proving the inequality for all integers $k \in \mathbb{N}^\times$, where $\mathbb{N}^\times = \mathbb{N} \setminus \{0\}$, as the case $k = 0$ does not correspond to a meaningful convex combination.

Base Case: For our Base Case, we will choose $k = 1$. In this case, $\lambda_1 = 1$ since $\lambda \in \Delta_1$ means the only possible weight is 1.

$$\text{Thus, } \sum_{i=1}^1 \lambda_i x_i = x_1 \text{ and } \sum_{i=1}^1 \lambda_i f(x_i) = f(x_1)$$

$$f\left(\sum_{i=1}^1 \lambda_i x_i\right) = f(x_1)$$

$$f(x_1) = \sum_{i=1}^1 \lambda_i f(x_i)$$

$$f\left(\sum_{i=1}^1 \lambda_i x_i\right) = \sum_{i=1}^1 \lambda_i f(x_i)$$

Hence, for the Base Case

$$f\left(\sum_{i=1}^1 \lambda_i x_i\right) \leq \sum_{i=1}^1 \lambda_i f(x_i)$$

Inductive Case and Step:

We have that $f(\lambda_1 x_1 + \lambda_2 x_2) \leq \lambda_1 f(x_1) + \lambda_2 f(x_2)$ holds by convexity.

Now, we assume that the inequality holds for $k = n$: $f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i)$.

Now, we consider for $k = n + 1$:

Let $\lambda' = \sum_{i=1}^n \lambda_i$. Because each $\lambda_i \geq 0$, we have that $\lambda' \geq 0$.

If $\lambda' = 0$, then $\lambda_{n+1} = 1$ and the inequality holds trivially. So, we will continue with $\lambda' > 0$.

$$\begin{aligned} \sum_{i=1}^{n+1} \lambda_i x_i &= \sum_{i=1}^n \lambda_i x_i + \lambda_{n+1} x_{n+1} \\ \sum_{i=1}^n \lambda_i x_i + \lambda_{n+1} x_{n+1} &= \lambda' \sum_{i=1}^n \frac{\lambda_i}{\lambda'} x_i + \lambda_{n+1} x_{n+1} \\ \sum_{i=1}^{n+1} \lambda_i x_i &= \lambda' \sum_{i=1}^n \frac{\lambda_i}{\lambda'} x_i + \lambda_{n+1} x_{n+1} \end{aligned}$$

$$f\left(\sum_{i=1}^{n+1} \lambda_i x_i\right) = f\left(\lambda' \sum_{i=1}^n \frac{\lambda_i}{\lambda'} x_i + \lambda_{n+1} x_{n+1}\right) \quad (\text{Rewriting equivalently})$$

$$f\left(\sum_{i=1}^{n+1} \lambda_i x_i\right) \leq \lambda' f\left(\sum_{i=1}^n \frac{\lambda_i}{\lambda'} x_i\right) + \lambda_{n+1} f(x_{n+1}) \quad (\text{Applying the convexity of } f)$$

$$f\left(\sum_{i=1}^n \frac{\lambda_i}{\lambda'} x_i\right) \leq \sum_{i=1}^n \frac{\lambda_i}{\lambda'} f(x_i)$$

$$f\left(\sum_{i=1}^n \frac{\lambda_i}{\lambda'} x_i\right) \leq \frac{1}{\lambda'} \sum_{i=1}^n \lambda_i f(x_i)$$

$$\lambda' f\left(\sum_{i=1}^n \frac{\lambda_i}{\lambda'} x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i)$$

$$\frac{\lambda'}{\lambda'} f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i)$$

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i)$$

Note that we can divide by λ' as we are continuing with the nontrivial case where $\lambda' > 0$ meaning that $\lambda' \neq 0$.

Thus, we have shown that

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i)$$

holds in all our cases as desired.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

Apply convexity of f :

$$f\left(\sum_{i=1}^{n+1} \lambda_i x_i\right) = f\left(\lambda' \sum_{i=1}^n \frac{\lambda_i}{\lambda'} x_i + \lambda_{n+1} x_{n+1}\right) \leq \frac{\lambda'}{2} f\left(\sum_{i=1}^n \frac{\lambda_i}{\lambda'} x_i\right) + \frac{\lambda_{n+1}}{2} f(x_{n+1})$$

Here, the mistake is that ChatGPT incorrectly applies convexity of f .

3. **Exercise 3.4:** Let $f_i : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ be convex functions indexed by an arbitrary set I . Show that the pointwise supremum function $f(x) = \sup_{i \in I} f_i(x)$ is convex.

A function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ is convex if and only if its epigraph $\text{epi}(f) = \{(x, r) \in \mathbb{E} \times \mathbb{R} : f(x) \leq r\}$ is a convex set in $\mathbb{E} \times \mathbb{R}$.

We continue by expressing $\text{epi}(f)$ in terms of $\text{epi}(f_i)$. By definition, the epigraph of f consists of all pairs (x, r) such that $f(x) \leq r$. Since $f(x)$ is defined as the supremum over all $f_i(x)$, the condition $f(x) \leq r$ is equivalent to requiring that $f_i(x) \leq r$ for every $i \in I$. In other words, a pair (x, r) belongs to the epigraph of f if and only if it belongs to the epigraph of each f_i . This shows that the epigraph of f is precisely the intersection of the epigraphs of f_i : $\text{epi}(f) = \bigcap_{i \in I} \text{epi}(f_i)$. Since each function f_i is convex,

its epigraph is a convex set, and because the intersection of any collection of convex sets is also convex, it follows that the epigraph of f is convex. Because there is a bijective correspondence between the convexity of a function and the convexity of its epigraph, it follows that f must be convex.

Each f_i is convex, so $\text{epi}(f_i)$ is convex. The intersection of an arbitrary family of convex sets is convex. Thus, because $\text{epi}(f)$ is convex, we have that $\bigcap_{i \in I} \text{epi}(f_i)$ is convex.

By the epigraph characterization, the pointwise supremum function $f(x) = \sup_{i \in I} f_i(x)$ is convex.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

A function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ is convex if and only if its epigraph

$$\text{epi}(f) = \{(x, r) \in \mathbb{E} \times \mathbb{R} : f(x) \leq r\}$$

is a convex subset of $\mathbb{E} \times \mathbb{R}$.

Now, for each $i \in I$, since f_i is convex, its epigraph $\text{epi}(f_i)$ is convex.

We claim that

$$\text{epi}(f) = \bigcup_{i \in I} \text{epi}(f_i)$$

Since each epigraph $\text{epi}(f_i)$ is convex, and the union of convex sets is convex, it follows that $\text{epi}(f)$ is convex.

Hence, f is convex.

The error here is that $\text{epi}(f) \neq \bigcup_{i \in I} \text{epi}(f_i)$ in general unlike what ChatGPT erroneously stated.

4. **Exercise 3.14:** Let $Q \subset \mathbb{E} \times \mathbb{R}$ be a set. Establish the following statements.

1. If the infimum in the definition of $E_Q(x)$ is attained at every point x where it is finite, then equality holds:

$$\text{epi}(E_Q) = Q + (\{0\} \times \mathbb{R}_+)$$

In order to prove that $\text{epi}(E_Q) = Q + (\{0\} \times \mathbb{R}_+)$, we will use double inclusion and prove that $\text{epi}(E_Q) \subseteq Q + (\{0\} \times \mathbb{R}_+)$ and $Q + (\{0\} \times \mathbb{R}_+) \subseteq \text{epi}(E_Q)$.

We will first show that $Q + (\{0\} \times \mathbb{R}_+) \subseteq \text{epi}(E_Q)$. Take any $(x, \alpha) \in Q + (\{0\} \times \mathbb{R}_+)$. By definition of the lower envelope, there exists $(x, r) \in Q$ and $s \in \mathbb{R}_+$, so that $\alpha = r + s$. Because, $E_Q = \inf\{r : (x, r) \in Q\}$, we have $E_Q(x) \leq r$. Thus, $\alpha = r + s$ from how we defined α , $r + s \geq r$ because $s \geq 0$, and $r \geq E_Q(x)$ as $E_Q(x)$ is an infimum of the set r such that $(x, r) \in Q$, and more relevantly, a lower bound here. From this, we can say $\alpha \geq E_Q(x)$. This means $(x, \alpha) \in \text{epi}(E_Q)$. Because (x, α) was chosen arbitrarily, we can say $Q + (\{0\} \times \mathbb{R}_+) \subseteq \text{epi}(E_Q)$.

Now, we will show that $\text{epi}(E_Q) \subseteq Q + (\{0\} \times \mathbb{R}_+)$. Suppose $(x, t) \in \text{epi}(E_Q)$. Then, by definition, $t \geq E_Q(x)$. If the infimum in the definition of $E_Q(x)$ is attained, then there exists $(x, r^*) \in Q$ such that $r^* = E_Q(x)$. We now set $s = t - r^*$ and we note that $s \geq 0$ as $t \geq r^*$ by definition of the lower envelope. We also have $(x, t) = (x, r^*) + (0, s)$ with $(x, r^*) \in Q$ and $s \in \mathbb{R}_+$ by applying the definition of Minkowski sum. Therefore, $(x, t) \in Q + (\{0\} \times \mathbb{R}_+)$. Because (x, t) was chosen arbitrarily in $\text{epi}(E_Q)$, we can say that $\text{epi}(E_Q) \subseteq Q + (\{0\} \times \mathbb{R}_+)$.

Because $Q + (\{0\} \times \mathbb{R}_+) \subseteq \text{epi}(E_Q)$ and $\text{epi}(E_Q) \subseteq Q + (\{0\} \times \mathbb{R}_+)$, we have that $\text{epi}(E_Q) = Q + (\{0\} \times \mathbb{R}_+)$.

What goes wrong if the infimum is not attained at some x with $E_Q(x)$ finite?

If the infimum is not attained at some x with $E_Q(x)$ finite, we have that $Q + (\{0\} \times \mathbb{R}_+) \subseteq \text{epi}(E_Q)$ holds, but $\text{epi}(E_Q) \not\subseteq Q + (\{0\} \times \mathbb{R}_+)$ in general. Consider the space $X = \{0\}$ with $Q = \{(0, t) \mid t > 0\}$. Then, $E_Q(0) = 0$ meaning that $\text{epi}(E_Q) = \{(0, t) \mid t \geq 0\}$. And this gives us, $Q + (\{0\} \times \mathbb{R}_+) = Q$ showing that without the infimum being attained at some x with $E_Q(x)$ finite, $\text{epi}(E_Q) \not\subseteq Q + (\{0\} \times \mathbb{R}_+)$ in general, or in other words, the inclusion does not follow.

2. If Q is convex, then the envelope $E_Q : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ is a convex function.

We know that $E_Q(x) = \inf\{r : (x, r) \in Q\}$ by definition. Furthermore, f is convex if for all $x \in \mathbb{E}$, $y \in \mathbb{E}$, and $\lambda \in [0, 1]$, $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$. Now, we will show that for any $x \in \mathbb{E}$, $y \in \mathbb{E}$, and $\lambda \in [0, 1]$, $E_Q(\lambda x + (1 - \lambda)y) \leq \lambda E_Q(x) + (1 - \lambda)E_Q(y)$.

Proof. Let $x \in \mathbb{E}$, $y \in \mathbb{E}$, and $\lambda \in [0, 1]$ be chosen arbitrarily.

For any $\epsilon > 0$, by the definition of infimum, there exist $r_x \in \mathbb{R}$ and $r_y \in \mathbb{R}$ such that

$$(x, r_x) \in Q, \quad r_x < E_Q(x) + \epsilon$$

$$(y, r_y) \in Q, \quad r_y < E_Q(y) + \epsilon$$

Since Q is convex, the convex combination

$$(\lambda x + (1 - \lambda)y, \lambda r_x + (1 - \lambda)r_y) \in Q$$

By the definition of E_Q ,

$$E_Q(\lambda x + (1 - \lambda)y) \leq \lambda r_x + (1 - \lambda)r_y$$

But $r_x < E_Q(x) + \epsilon$ and $r_y < E_Q(y) + \epsilon$, so

$$\lambda r_x + (1 - \lambda)r_y < \lambda E_Q(x) + (1 - \lambda)E_Q(y) + \epsilon$$

Therefore,

$$E_Q(\lambda x + (1 - \lambda)y) < \lambda E_Q(x) + (1 - \lambda)E_Q(y) + \epsilon$$

Since $\epsilon > 0$ was arbitrary, it follows that

$$E_Q(\lambda x + (1 - \lambda)y) \leq \lambda E_Q(x) + (1 - \lambda)E_Q(y)$$

Thus, E_Q is convex. □

Acknowledgment: For this question, I referenced Convex Analysis by R T Rockafellar, Variational Analysis by R T Rockafellar, and Math Stack Exchange.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

2. If Q is convex, then E_Q is convex.

Let $x, y \in \mathbb{E}$, $\lambda \in [0, 1]$, and fix any $\epsilon > 0$.

By the definition of infimum, there exist $r_x < E_Q(x) + \epsilon$ and $r_y < E_Q(y) + \epsilon$ such that:

$$(x, r_x), (y, r_y) \in Q$$

Since Q is convex:

$$\begin{aligned} (\lambda x + (1 - \lambda)y, \lambda r_x + (1 - \lambda)r_y) &\in Q \Rightarrow E_Q(\lambda x + (1 - \lambda)y) < \lambda r_x + (1 - \lambda)r_y \\ &< \lambda E_Q(x) + (1 - \lambda)E_Q(y) + \epsilon \Rightarrow E_Q(\lambda x + (1 - \lambda)y) < \lambda E_Q(x) + (1 - \lambda)E_Q(y) + \epsilon \end{aligned}$$

Taking the limit as $\epsilon \rightarrow 0$, we conclude:

$$E_Q(\lambda x + (1 - \lambda)y) \leq \lambda E_Q(x) + (1 - \lambda)E_Q(y)$$

Hence, E_Q is convex.

Here, the error is that we are not supposed to be taking the limit as $\epsilon \rightarrow 0$. Rather, we should try to prove that this inequality holds for any $\epsilon > 0$.

5. **Exercise 3.24:** Show that for any affine function $f(x) = \langle a, x \rangle + b$, the conjugate takes the form $f^*(y) = -b + \delta_{\{a\}}(y)$. Deduce the equality $f^{**} = f$.

We have

$$\begin{aligned}
 f^*(y) &= \sup_{x \in \mathbb{E}} \{ \langle y, x \rangle - f(x) \} \\
 &= \sup_{x \in \mathbb{E}} \{ \langle y, x \rangle - (\langle a, x \rangle + b) \} \\
 &= \sup_{x \in \mathbb{E}} \{ \langle y - a, x \rangle - b \} \\
 &= \sup_{x \in \mathbb{E}} \langle y - a, x \rangle - b
 \end{aligned}$$

So, $f^*(y) = \sup_{x \in \mathbb{E}} \langle y - a, x \rangle - b$. If $y \neq a$, the function $x \mapsto \langle y - a, x \rangle$ is unbounded above as we just pick x in the direction of $y - a$, so the supremum is just ∞ . However, if $y = a$, then $\langle y - a, x \rangle = 0$ for all x , so the supremum is 0. So,

$$\begin{aligned}
 f^*(y) &= \begin{cases} -b & \text{if } y = a \\ \infty & \text{if } y \neq a \end{cases} \\
 f^*(y) &= -b + \delta_{\{a\}}(y)
 \end{aligned}$$

By **Definition 3.22**, the Fenchel conjugate f^* is

$$f^*(y) = \sup_{x \in \mathbb{E}} \{ \langle y, x \rangle - f(x) \}$$

We can further say

$$f^{**}(y) = \sup_{x \in \mathbb{E}} \{ \langle y, x \rangle - f^*(x) \}$$

We know that

$$\begin{aligned}
 f^*(y) &= -b + \delta_{\{a\}}(y) \\
 \langle x, y \rangle - f^*(y) &= \begin{cases} \langle x, y \rangle + b & \text{if } y = a \\ -\infty & \text{if } y \neq a \end{cases}
 \end{aligned}$$

Thus, the supremum is only attained at $y = a$:

$$f^{**} = \langle x, a \rangle + b$$

$$f^{**}(x) = \langle a, x \rangle + b \quad (\text{By symmetry of the inner product})$$

$$f^{**}(x) = f(x) \quad (\text{For any affine function } f(x) = \langle a, x \rangle + b)$$

We have shown that for any affine function $f(x) = \langle a, x \rangle + b$, $f^{**}(x) = f(x)$.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

Let $f(x) = \langle a, x \rangle + b$ be an affine function on $\mathbb{E}$. We compute the Fenchel conjugate:

$$f^*(y) = \sup_{x \in \mathbb{E}} \{ \langle y, x \rangle - f(x) \} = \sup_{x \in \mathbb{E}} \{ \langle y - a, x \rangle - b \} = \sup_{x \in \mathbb{E}} \langle y - a, x \rangle - b$$

We now compute this supremum. Observe that:

- If $y = a$, then $\langle y - a, x \rangle = 0$, so $f^*(y) = -b$
- If $y \neq a$, then $\langle y - a, x \rangle$ is bounded above, so the supremum is finite.

Therefore,

$$f^*(y) = \begin{cases} -b & \text{if } y = a \\ \sup_{x \in \mathbb{E}} \langle y - a, x \rangle - b & \text{if } y \neq a \end{cases} \Rightarrow f^*(y) = -b + \delta_{\{a\}}(y)$$

Now compute the biconjugate:

$$f^{**}(x) = \inf_{y \in \mathbb{E}} \{ f^*(y) - \langle x, y \rangle \}$$

We plug in:

$$f^*(y) = -b + \delta_{\{a\}}(y) \Rightarrow f^{**}(x) = \inf_{y \in \mathbb{E}} \left\{ \begin{cases} -b - \langle x, y \rangle & \text{if } y = a \\ \infty & \text{otherwise} \end{cases} \right\} = -b - \langle x, a \rangle$$

Hence, $f^{**}(x) = -b - \langle x, a \rangle$.

Here, ChatGPT makes an error by stating that $f^{**}(x) = \inf_{y \in \mathbb{E}} \{ f^*(y) - \langle x, y \rangle \}$. The problem, here, is that we should be using the definition of supremum or suprema, not infimum or infima. Rather, $f^{**}(x) = \sup_{y \in \mathbb{E}} \{ \langle x, y \rangle - f^*(y) \}$.

6. **Exercise 3.25:** For any function $g : \mathbb{E} \times \mathbb{Y} \rightarrow \overline{\mathbb{R}}$, the estimate holds:

$$\sup_y \inf_x g(x, y) \leq \inf_x \sup_y g(x, y)$$

Obviously, a lower bound for a function cannot be greater than an upper bound for the same function. Thus, because the infimum is a lower bound and the supremum is an upper bound, $\inf_x g(x, y) \leq \sup_y g(x, y)$, where x and y are chosen arbitrarily with $x \in \mathbb{E}$ and $y \in \mathbb{Y}$.

Now, we take the supremum over y on $\inf_x g(x, y)$ to get: $\sup_y \inf_x g(x, y) \leq \sup_y g(x, y)$.

Similarly, we take the infimum over x on $\sup_y g(x, y)$ to get: $\sup_y \inf_x g(x, y) \leq \inf_x \sup_y g(x, y)$.

Thus,

$$\sup_y \inf_x g(x, y) \leq \inf_x \sup_y g(x, y)$$

Acknowledgment: For this question, I got help from Batool.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

Suppose that $g(x, y)$ is convex in x and concave in y . Then by Sion's minimax theorem, we know:

$$\sup_y \inf_x g(x, y) = \inf_x \sup_y g(x, y)$$

Therefore, the inequality holds as a special case of this equality.

Since this is true when g is convex-concave, it must also hold when g is arbitrary, because removing structure can only reduce the gap between the sup and inf.

Thus:

$$\sup_y \inf_x g(x, y) \leq \inf_x \sup_y g(x, y)$$

holds in general.

Here, ChatGPT makes an error: Sion's minimax theorem is not applicable here and is applied incorrectly when it is applied in the context of suprema and infima.

7. **Exercise 3.34:** Consider two proper functions $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ and $g : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ and suppose that f is differentiable at a point x . Show the equality

$$\partial(f + g)(x) = \nabla f(x) + \partial g(x)$$

We will prove this by Double Inclusion.

We will first show that $\partial(f + g)(x) \subseteq \nabla f(x) + \partial g(x)$. Let $s \in \partial(f + g)(x)$.

$$(f + g)(y) \geq (f + g)(x) + \langle s, y - x \rangle \quad \forall y \in \mathbb{E} \quad (\text{By definition})$$

$$f(y) = f(x) + \langle \nabla f(x), y - x \rangle + o(\|y - x\|) \quad (\text{As } f \text{ is differentiable at } x)$$

$$f(y) + g(y) \geq f(x) + g(x) + \langle s, y - x \rangle$$

$$g(y) \geq g(x) + \langle s, y - x \rangle - f(y) + f(x)$$

$$g(y) \geq g(x) + \langle s, y - x \rangle - (f(y) - f(x))$$

$$f(y) - f(x) = \langle \nabla f(x), y - x \rangle + o(\|y - x\|)$$

$$g(y) \geq g(x) + \langle s, y - x \rangle - (\langle \nabla f(x), y - x \rangle + o(\|y - x\|))$$

$$g(y) \geq g(x) + \langle s, y - x \rangle - \langle \nabla f(x), y - x \rangle - o(\|y - x\|)$$

$$\text{As } y \rightarrow x, o(\|y - x\|) \rightarrow 0$$

$$g(y) \geq g(x) + \langle s - \nabla f(x), y - x \rangle \quad \forall y \in \mathbb{E}$$

This means $s - \nabla f(x) \in \partial g(x)$, so $s \in \nabla f(x) + \partial g(x)$. Thus, $\partial(f + g)(x) \subseteq \nabla f(x) + \partial g(x)$.

We will second show that $\nabla f(x) + \partial g(x) \subseteq \partial(f + g)(x)$. Let $t \in \partial g(x)$. This means $\nabla f(x) + t \in \partial(f + g)(x)$.

$$g(y) \geq g(x) + \langle t, y - x \rangle, \quad f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle, \quad \forall y \in \mathbb{E}$$

$$(f + g)(y) \geq (f + g)(x) + \langle \nabla f(x) + t, y - x \rangle$$

$$\nabla f(x) + t \in \partial(f + g)(x)$$

$$\nabla f(x) + \partial g(x) \subseteq \partial(f + g)(x)$$

Because $\partial(f + g)(x) \subseteq \nabla f(x) + \partial g(x)$ and $\nabla f(x) + \partial g(x) \subseteq \partial(f + g)(x)$, we have that $\partial(f + g)(x) = \nabla f(x) + \partial g(x)$.

Why the equality $\partial\delta_Q(x) = N_Q(x)$ holds: This equality holds because when $x \notin Q$, $\delta_Q(x) = \infty$ meaning that both $\partial\delta_Q(x) = \emptyset$ and $N_Q(x) = \emptyset$ meaning that $\partial\delta_Q(x) = N_Q(x)$ and when $x \in Q$, we have that $\partial\delta_Q(x) = \{v \mid \langle v, y - x \rangle \leq 0 \quad \forall y \in Q\}$, which is basically the definition of $N_Q(x)$.

Acknowledgment: For this question, I got help from Batool.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

"Since g is convex, its subdifferential $\partial g(x)$ is always a singleton set, meaning that $\partial g(x) = \{\nabla g(x)\}$, even if g is not differentiable."

Here, the error is that $\partial g(x) \neq \{\nabla g(x)\}$ and g being differentiable is a noticeable factor here that seems to be ignored.

8. **Exercise 3.36:** Consider a convex function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ and a point x with $f(x)$ finite. Show that the inclusion $v \in \partial f(x)$ holds if and only if

$$f(y) \geq f(x) + \langle v, y - x \rangle \text{ holds for all } y \in \mathbb{E}$$

If $v \in \partial f(x)$, then by the definition of the subgradient, $f(y) \geq f(x) + \langle v, y - x \rangle$ holds for all $y \in \mathbb{E}$. If $f(y) \geq f(x) + \langle v, y - x \rangle$ holds for all $y \in \mathbb{E}$, $v \in \partial f(x)$ by the definition of the subgradient. Because if $v \in \partial f(x)$, then $f(y) \geq f(x) + \langle v, y - x \rangle$ holds for all $y \in \mathbb{E}$ and if $f(y) \geq f(x) + \langle v, y - x \rangle$ holds for all $y \in \mathbb{E}$, then $v \in \partial f(x)$, we have that $v \in \partial f(x)$ holds if and only if $f(y) \geq f(x) + \langle v, y - x \rangle$ holds for all $y \in \mathbb{E}$.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

Suppose $v \in \partial f(x)$. Then by definition:

$$f(y) \geq f(x) + \langle v, y - x \rangle \quad \forall y \in \mathbb{E}$$

Now, since all convex functions are affine wherever they are finite, this inequality becomes an equality:

$$f(y) = f(x) + \langle v, y - x \rangle$$

for all $y \in \mathbb{E}$, so f is linear on its domain.

Conversely, suppose the inequality holds for all y . Then by definition, $v \in \partial f(x)$.

Hence,

$$v \in \partial f(x) \iff f(y) = f(x) + \langle v, y - x \rangle \quad \forall y \in \mathbb{E}$$

Here, ChatGPT makes two errors: all convex are not affine wherever they are finite contrary to what ChatGPT stated and $f(y) \neq f(x) + \langle v, y - x \rangle$ in general. Rather, $f(y) \geq f(x) + \langle v, y - x \rangle$ for all $y \in \mathbb{E}$.

9. **Exercise 3.58:** Any proper, closed, μ strongly convex function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ has a unique minimizer x , which moreover satisfies

$$f(y) - f(x) \geq \frac{\mu}{2} \|y - x\|^2 \quad \text{for all } y \in \mathbb{E}$$

Here, f is coercive because as $\|x\| \rightarrow \infty$, $f(x) \rightarrow \infty$.

Because f is coercive, there exists a minimizer x .

We have $v \in \partial f(x)$ and because $v = 0$, $0 \in \partial f(x)$.

$$f(y) \geq f(x) + \underbrace{\langle v, y - x \rangle}_0 + \frac{\mu}{2} \|y - x\|^2$$

$$f(y) \geq f(x) + \frac{\mu}{2} \|y - x\|^2$$

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

Since f is strongly convex and proper, it is coercive, so there exists a unique minimizer x such that:

$$f(x) = 0$$

Then for all $y \in \mathbb{E}$, we use the strong convexity inequality:

$$f(y) \geq f(x) + \frac{\mu}{2} \|y - x\|^2 = \frac{\mu}{2} \|y - x\|^2$$

Hence, the function always satisfies:

$$f(y) \geq \frac{\mu}{2} \|y - x\|^2$$

Here, ChatGPT makes a mistake: Though $f(x)$ might possibly be 0 somewhere for a unique minimizer x , this does not apply in general.